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Mathematics

Probability

IDENTIFICATION

CODE :	IF-3-PROB
ECTS :	2.0

HOURS

Lectures :	7.5 h
Seminars :	16.0 h
Laboratory :	8.0 h
Project :	0.0 h
Teacher-student contact :	31.5 h
Personal work :	20.0 h
Total :	51.5 h

ASSESSMENT METHOD

Exam (1h30, all documents authorized).
Evaluation of a report on practical work, at the end of the sessions.

TEACHING AIDS

Lecture notes [1].
Description of the practical work.

TEACHING LANGUAGE

French

CONTACT

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AIMS

To acquire the basics of probabilistic tools, to be able to apply them to stochastic modeling and statistics.

Targeted skills:
- Description of phenomena using a probabilistic model: random variables and stochastic distributions.
- Extraction of some relevant features from a model.
- Understanding of law of large numbers and central limit theorem and their repercussions
- Representations of a dynamical system using graphs, matrices and probabilistic model: Markov chains.

CONTENT

Outline:
1- Recalling - Counting
2- Bases of probability theory
3- Random variables
4- Random vectors
5- Limit theorems
6- Introduction to Markov chains

Practical sessions:
- random variables simulations
- test of the quality of simulation
- queuing simulation

BIBLIOGRAPHY

[1] Mazet O. Cours de probabilités 3IF 2005-2006. Disponible sur http://www-gmm.insa-toulouse.fr/~omazet/Enseignement/Cours/Cours_Proba.pdf
[2] Saporta G. (1990) Probabilités, analyse de données et statistique. Paris : Ed. Technip

PRE-REQUISITE

Undergraduate level is needed for : theory of integration, numerical sequences and series, and matrices